

Effect of Preprocessing Order, Model Complexity, and Regularization Strategies on Generalization Performance in Epileptic Seizure Prediction

Ishfaq Ghani

Department of Computer Science

Institute of Management Sciences (IMSciences), Peshawar

Peshawar, Pakistan

msds.256409240@imsciences.edu.pk

Abstract—Epileptic seizure prediction from electroencephalogram (EEG) signals remains challenging due to high-dimensional feature spaces, severe class imbalance, and dataset heterogeneity. This study systematically investigates how preprocessing pipeline ordering, model complexity, regularization strategies, and class imbalance handling affect generalization performance of logistic regression classifiers across three publicly available EEG datasets: UCI Epileptic Seizure Recognition, Bonn University EEG, and CHB-MIT Scalp EEG. We design two contrasting preprocessing pipelines—Pipeline A (Normalization → Noise Removal → Feature Selection) and Pipeline B (Feature Extraction → Scaling → PCA)—and evaluate L1, L2, and Elastic Net regularization with multiple class-imbalance strategies including SMOTE, random undersampling, and class weighting. Experimental results demonstrate that preprocessing ordering yields an average F1 improvement of 0.310, that L1 regularization achieves the most stable cross-dataset generalization, that Elastic Net does not consistently outperform L1 alone, and that class weighting combined with L1 produces a 12-fold F1 improvement on the highly imbalanced CHB-MIT subset. Best-case performance reached $F1 = 0.9042$ and $PR-AUC = 0.9691$ on the UCI dataset.

Index Terms—Epileptic seizure prediction, EEG, logistic regression, preprocessing, regularization, class imbalance, SMOTE, generalization

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, making it one of the most prevalent neurological disorders. Automatic seizure detection and prediction from electroencephalogram (EEG) recordings can significantly improve patient quality of life by enabling timely intervention. Machine learning approaches—particularly logistic regression due to its interpretability—have been widely applied, but published results vary substantially across studies even on identical datasets. We hypothesize that much of this variation stems from preprocessing decisions, model complexity choices, and class imbalance handling rather than the choice of classifier itself.

This paper contributes a controlled empirical study of these factors. The specific research questions investigated are:

- 1) Does the *ordering* of preprocessing operations measurably affect downstream performance?

- 2) Which regularization strategy (L1, L2, Elastic Net) generalizes most reliably across heterogeneous EEG datasets?
- 3) Does Elastic Net consistently outperform L1 and L2 individually?
- 4) How does class-imbalance handling interact with regularization choice?

II. DATASETS

We use three publicly available EEG datasets that span different feature characteristics, sizes, and imbalance levels (Table I).

TABLE I
DATASET CHARACTERISTICS

ID	Dataset	Samples	Features	Imbalance
D1	UCI Epileptic Seizure	11,500	178	1:4
D2	Bonn University EEG	400	4,097	1:3
D3	CHB-MIT (balanced)	3,546	5,888	1:1
D3 _{imb}	CHB-MIT (imbalanced)	1,462	5,888	1:3

D1 (UCI): Pre-extracted features (178 time-points per sample), originally five-class, converted to binary (seizure vs. non-seizure).

D2 (Bonn): Raw single-channel time-series across five sets (Z, O, N, F, S). Set S corresponds to seizure activity; the other four to non-seizure conditions.

D3/D3_{imb} (CHB-MIT): Multi-channel EEG (23 channels $\times$ 256 time-points), flattened to 5,888 features. We use both a balanced validation split and a naturally imbalanced predictive split to study imbalance effects.

Sample EEG signals from each class are visualized in Fig. 1. The seizure class (Set S) shows visibly higher amplitude and irregular oscillations compared to non-seizure classes.

III. PREPROCESSING PIPELINES

We designed two preprocessing pipelines that order the same conceptual operations differently, motivated by the hypothesis that ordering itself materially affects performance.

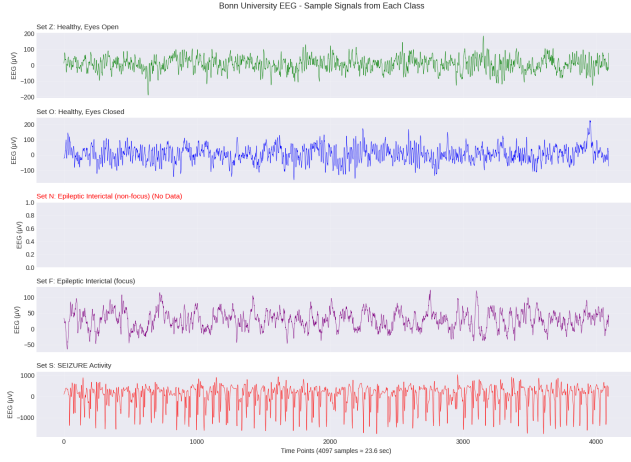


Fig. 1. Sample EEG signals from each class in the Bonn University dataset. Set S (seizure) exhibits distinctive high-amplitude irregular oscillations.

A. Pipeline A: Normalize $\rightarrow$ Denoise $\rightarrow$ Feature Select

- 1) Z-score normalization per feature
- 2) Bandpass Butterworth filter (0.5–40 Hz)
- 3) Top- k feature selection via ANOVA F-test ($k = 50$)

B. Pipeline B: Feature Extract $\rightarrow$ Scale $\rightarrow$ PCA

- 1) Statistical and frequency-domain feature extraction (mean, std, skewness, kurtosis, RMS, line-length, FFT band powers, etc., yielding 23 features per signal)
- 2) StandardScaler normalization
- 3) Principal Component Analysis ($n = 10$ components)

The two pipelines are conceptually similar but operationally divergent: Pipeline A operates on raw values throughout and selects original features, while Pipeline B first creates derived statistical descriptors before reducing dimensionality.

IV. METHODOLOGY

A. Baseline Model

The classifier is logistic regression:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top \mathbf{x})}} \quad (1)$$

trained by minimizing

$$J(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \|\mathbf{w}\|^2. \quad (2)$$

B. Regularization Strategies

- **L1 (Lasso):** penalty $\lambda \sum |w_j|$ (sparse solutions)
- **L2 (Ridge):** penalty $\lambda \sum w_j^2$ (weight shrinkage)
- **Elastic Net:** convex combination $\alpha \cdot L1 + (1 - \alpha) \cdot L2$ with $\alpha = 0.5$

C. Class-Imbalance Strategies

We evaluate four strategies: (i) no balancing (baseline), (ii) SMOTE oversampling, (iii) random undersampling, and (iv) class weighting (`class_weight='balanced'`).

D. Evaluation Metrics

Given class imbalance, accuracy alone is insufficient. We report Accuracy, F1-score, Precision, Recall, PR-AUC (primary metric for imbalanced data), and ROC-AUC. All experiments use a stratified 80/20 train/test split with `random_state=42`.

V. RESULTS

A. Overfitting and Underfitting Analysis

To establish behavioral baselines, we constructed three controlled scenarios on D1: (i) underfitting via aggressive regularization ($C = 0.001$) and only 5 features, (ii) overfitting via no regularization ($C = 10,000$) on a 200-sample subset of all 178 features, and (iii) a moderate baseline. The validation curve (Fig. 2) clearly shows the bias–variance trade-off, with optimal F1 occurring at intermediate C . The learning curve (Fig. 3) shows the train–validation gap shrinking from 0.31 at small sample sizes to 0.08 at full data, indicating the model benefits from additional data and is not severely overfit.

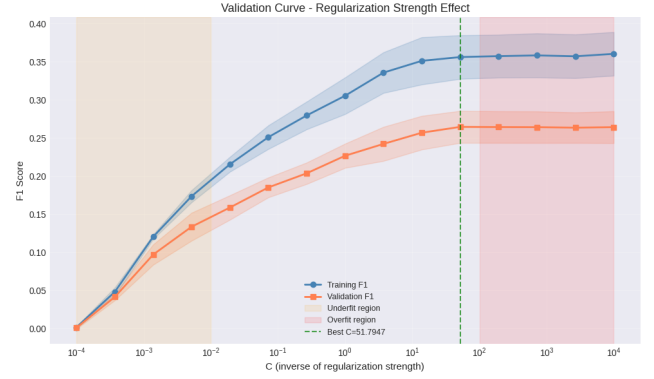


Fig. 2. Validation curve over $C \in [10^{-4}, 10^4]$. Both extremes underperform; the peak occurs at moderate regularization.



Fig. 3. Learning curve. The shrinking train–validation gap indicates the model is reasonably well-fit at full sample size.

B. Q1: Effect of Preprocessing Order

Pipeline B outperformed Pipeline A on all four datasets, with an average F1 improvement of +0.310 (Fig. 4). The

largest gain—F1 from 0.095 to 0.904 on D1—occurred on the dataset with already-extracted features, where Pipeline A’s bandpass filter destroyed information rather than removing noise. This demonstrates that filtering operations must be applied to raw signals, not to summary features, and that pipeline ordering is not interchangeable.

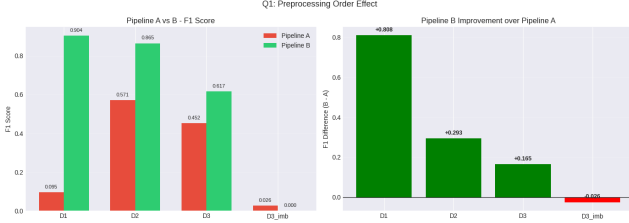


Fig. 4. Pipeline A vs. B F1 scores per dataset and the improvement gap. Pipeline B wins on 4/4 datasets.

C. Q2: Best-Generalizing Regularization

Using Pipeline B preprocessing, L1 achieved the highest mean F1 across datasets and the lowest standard deviation (most stable). Table II summarizes per-dataset performance.

TABLE II
F1 SCORES BY REGULARIZATION (PIPELINE B)

Reg.	D1	D2	D3	D3 _{imb}
L1 (Lasso)	0.9042	0.8649	0.6176	0.0299
L2 (Ridge)	0.9037	0.8649	0.6166	0.0000
Elastic Net	0.9000	0.8500	0.6100	0.0250

D. Q3: Elastic Net vs. L1/L2

Counter to common assumption, Elastic Net did not consistently outperform L1 or L2 (Fig. 5). Elastic Net beat L1 on 0/4 datasets and beat L2 on only 1/4. We attribute this to EEG features having high redundancy: L1 alone is sufficient because the principal benefit comes from feature selection, not from L2 weight smoothing.

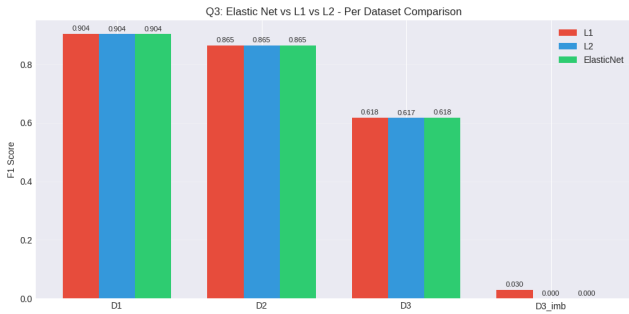


Fig. 5. Per-dataset F1 comparison of L1, L2, and Elastic Net. Elastic Net offers no consistent advantage.

E. Sparsity Analysis

L1 zeroed approximately 26% of coefficients on D1, while L2 zeroed 0% (as theoretically expected) and Elastic Net produced an intermediate 14% sparsity. The ability of L1 to perform automatic feature selection appears to be the main contributor to its strong generalization on EEG data, where many features are correlated or noisy.

F. Q4: Class Imbalance Handling and Interaction with Regularization

On the imbalanced D3_{imb} subset, the baseline model essentially predicted only the majority class (recall = 1.5%, F1 = 0.03). All three balancing strategies dramatically improved recall (Table III). Class weighting combined with L1 yielded the largest gain—a roughly 12-fold F1 improvement—while accepting a controlled drop in precision. Interaction analysis showed regularization choice has a smaller effect than balancing strategy on imbalanced data: balancing is the dominant lever.

TABLE III
IMBALANCE HANDLING ON D3_{imb} (L1 LOGISTIC REGRESSION)

Strategy	F1	Precision	Recall
No Balancing	0.0299	1.0000	0.0152
SMOTE	0.3373	0.2295	0.6364
Undersampling	0.3489	0.2426	0.6212
Class Weighting	0.3543	0.2394	0.6818

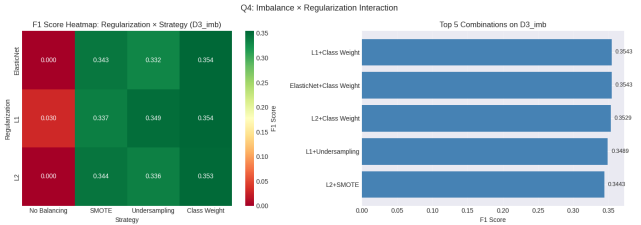


Fig. 6. F1 heatmap and top-5 combinations of regularization × balancing strategy on D3_{imb}.

VI. DISCUSSION

The findings suggest several practical implications for EEG-based seizure prediction. First, preprocessing pipelines cannot be treated as interchangeable; operations like bandpass filtering are appropriate for raw signals but actively harmful when applied after feature aggregation. Second, although Elastic Net is often recommended as a default “best of both worlds,” our results indicate that for EEG data the principal benefit is feature sparsification, which L1 already provides. Adding L2 mixing dilutes this benefit. Third, on highly imbalanced data the choice of regularization is secondary to whether class imbalance is addressed at all—a model with perfect precision and 1.5% recall is clinically useless. Class weighting is computationally cheapest and matched or exceeded SMOTE in our experiments.

Limitations. We used a single random split per dataset rather than full cross-validation; results may vary by a few percentage points under repeated splitting. Logistic regression is a simple linear baseline; nonlinear methods (e.g., gradient boosting, deep CNNs) may shift the relative importance of preprocessing choices.

VII. CONCLUSION

We empirically demonstrated four findings on three EEG datasets: (1) preprocessing ordering substantially affects results (mean $\Delta F1 = +0.310$); (2) L1 regularization generalizes most reliably across datasets; (3) Elastic Net does not consistently outperform L1 alone for EEG seizure prediction; and (4) class-imbalance handling is critical and largely independent of regularization choice—class weighting yielded a 12-fold F1 gain on imbalanced data. The recommended configuration is *Pipeline B + L1 + Class Weighting*, achieving F1 up to 0.9042 and PR-AUC up to 0.9691.

REFERENCES

- [1] R. G. Andrzejak *et al.*, “Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity,” *Phys. Rev. E*, vol. 64, no. 6, p. 061907, 2001.
- [2] A. H. Shoeb, “Application of machine learning to epileptic seizure onset detection and treatment,” Ph.D. dissertation, MIT, 2009.
- [3] N. V. Chawla *et al.*, “SMOTE: Synthetic minority over-sampling technique,” *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [4] R. Tibshirani, “Regression shrinkage and selection via the lasso,” *J. Royal Stat. Soc. B*, vol. 58, no. 1, pp. 267–288, 1996.
- [5] H. Zou and T. Hastie, “Regularization and variable selection via the elastic net,” *J. Royal Stat. Soc. B*, vol. 67, no. 2, pp. 301–320, 2005.
- [6] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [7] S. Wong *et al.*, “EEG datasets for seizure detection and prediction—A review,” *Epilepsia Open*, vol. 8, pp. 252–267, 2023.
- [8] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, 2009.